Work Done So Far

PPE Compliance Detection • 30 Oct 2026 • Overall about **80% complete**

1. The project in one paragraph

Supervisors cannot watch every worker. Our system takes a photo (or short video), finds every worker, and checks **that specific worker** for a helmet and a vest. It draws a green, red or amber box per worker with a label such as "Worker 2: NO HELMET", and produces a violation summary you can download. A helmet on a table or in someone's hand does not count, because we match each item to the person wearing it.

2. Everything we did, step by step

#	Work	What we did	Who did the thinking
1	**Research**	Studied open-source PPE projects and public datasets for *ideas only* (nothing copied). Found that none of them cleanly matched each helmet to the specific worker wearing it, so we built that ourselves.	Team + Claude
2	**Set-up**	Python 3.12 environment, GPU-enabled PyTorch on the laptop's RTX 5060, and the project folder with clean modules.	Claude, team reviewed
3	**Data, round 1**	Downloaded the public *Construction-PPE* dataset (1,416 photos) and reduced it to three classes: person, helmet, vest.	Claude, team reviewed
4	**Training, round 1**	Fine-tuned a pretrained YOLO11s model: 60 epochs, 36 minutes. Scored on photos it never saw.	Claude, team reviewed
5	**Matching logic (the core)**	Each worker gets a head zone and a torso zone. A helmet counts only in the head zone, a vest only on the torso, and each item can belong to at most one worker. 69 automatic tests check it with hand-made boxes.	Our own design
6	**Rules and verdicts**	Helmet and vest required (configurable). Three outcomes: COMPLIANT, VIOLATION, NEEDS_REVIEW, each with a one-sentence reason. Tiny, cut-off, crowded or borderline workers go to REVIEW instead of a false accusation.	Our own design
7	**Hard conditions**	Dark photos are brightened automatically (CLAHE). Larger image size and zoomed-in tiles help find small, far-away workers. Crowds get readable, non-overlapping labels.	Our own design
8	**Outputs**	Annotated image, per-worker table, violation summary, CSV and JSON download, a command-line tool, and a batch analyser that processes a whole folder into an HTML report.	Our own work
9	**Web app**	Streamlit app: upload photos, see results and the reason for every decision, download files. A second page handles video.	Our own work

10	**Video**	Each worker is tracked through the clip, and a violation is reported only after it persists for 5 of the last 8 usable frames, with an evidence photo. Tested on synthetic video; **not yet on a real clip**.	Our own work
11	**Data, round 2**	Added the public *CHVG* dataset (1,699 more photos, CC BY 4.0), converted from XML, merged into 2,491 training images, with duplicate photos kept in the same split so tests stay fair.	Claude, team reviewed
12	**Training, round 2**	Fine-tuned our round-1 model on the merged data: 30 epochs, 40 minutes. It became the default model after it did better on the team's real site photos.	Claude, team reviewed
13	**Fixing real problems**	Found from real photos: a helmet seen with low confidence caused a false violation (now REVIEW); the same helmet detected twice appeared as "unassigned" (duplicates merged); background workers were missed (zoomed tiling added).	Team spotted, Claude fixed
14	**Testing**	69 automatic tests pass. An evaluation script scores any photo set against an answer key by hard-case group.	Our own work
15	**Documents**	README with credits, walkthrough of every module, 3-minute demo script, team guide, and progress PDFs.	Claude, team reviewed

3. Our results (say these exactly)

Measured on photos never used for training (mAP@50)	Round 1	Round 2
Original test photos (141) — same source as the training photos	0.866	0.829
CHVG test photos (173) — from other sites	0.758	0.882*

*Round 2 trained on the training half of this collection, so it is not a like-for-like comparison. On the original photos round 2 is slightly lower. On the team's four real site photos, round 2 found more workers and fixed two wrong verdicts.

4. What is done and what is not

Done and working	Not done yet (be upfront about it)
✓ Detect people, helmets, vests ✓ Match each item to the right worker ✓ Rules per worker, flag the missing item ✓ Annotated images with a verdict per worker ✓ Violation summary and CSV/JSON export ✓ Multiple workers, occlusion (REVIEW), dark photos, small workers ✓ Web app, batch analyser, README, 69 tests	✗ Measured accuracy on the hard cases (helmet in hand, on a table, hi-vis non-vest, dark, distant): the test script is ready but the answer key and photos are not ✗ Video tested on a real clip ◐ Hi-vis shirts: the model counts them as vests; the team has not decided the rule ◐ Small background workers: improved, not perfect; a duplicate box can still appear ◐ Life jackets and similar look-alike clothing can be mistaken for vests

5. Explain it to the judges in 90 seconds

> "We built a system that checks each worker for a helmet and a vest. The key idea is that we don't just look for a helmet somewhere in the picture; we check whether it is on **that worker's head**, and each helmet can only belong to one worker. So a helmet on a table or in someone's hand doesn't clear anyone.
>
> For detection we fine-tuned a pretrained YOLO model on two public datasets, which is why training took under an hour on a laptop GPU. The matching logic, rules, summary, web app and tests are our own work. When we can't judge a worker fairly, because they are too small, cut off or crowded, we say 'needs review' instead of accusing them.
>
> We measured it on photos the model never saw. On photos from other sites round 1 scored 0.76, so we added a second dataset and a second round of training. We found real problems on our own site photos, like missed background workers and false alarms, and fixed them. We still have limits: small background workers and look-alike clothing such as life jackets can fool it, and we haven't yet measured the hard cases on a full test set of our own. That is our next step."

6. If a judge asks you personally

Question	Safe answer
What did *you* work on?	Pick your role from the Team Guide (data and training, matching logic, rules and app, or testing and results) and explain that part in two sentences.
Did you train it from scratch?	No. We fine-tuned a pretrained model. That is why it was fast.
What is your accuracy?	Give both numbers from section 3 and say the other-sites number is the more realistic one.
Where does it fail?	Small background workers, look-alike clothing like life jackets, and seated or bending workers who may become REVIEW.
Who wrote the code?	The matching, rules, summary, app and tests are our own. We used YOLO and public datasets for detection, and Claude as a coding assistant, which we reviewed and understand. It is credited in the README.

> **Never say:** "trained from scratch", "we collected the data", "100% accurate", or that it handles every hard case. Each of those is untrue or unproven, and one follow-up question would expose it.